VAE

VAE Space Analysis

Sunghyun Park

YONSEI UNIVERSITY

Variational Autoencoder (Loss)

How to maximize $\log p(x)$? Maximizing ELBO (A) + (B) = ELBO

$$\log p_\theta(x) = \underbrace{\mathbb{E}_z[\log p_\theta(x|z)]}_{\text{(A)}} - \underbrace{\mathrm{KL}\big(q_\phi(z|x)\|p(z)\big)}_{\text{(B)}} + \underbrace{\mathrm{KL}\big(q_\phi(z|x)\|p_\theta(z|x)\big)}_{\text{(C)}}$$

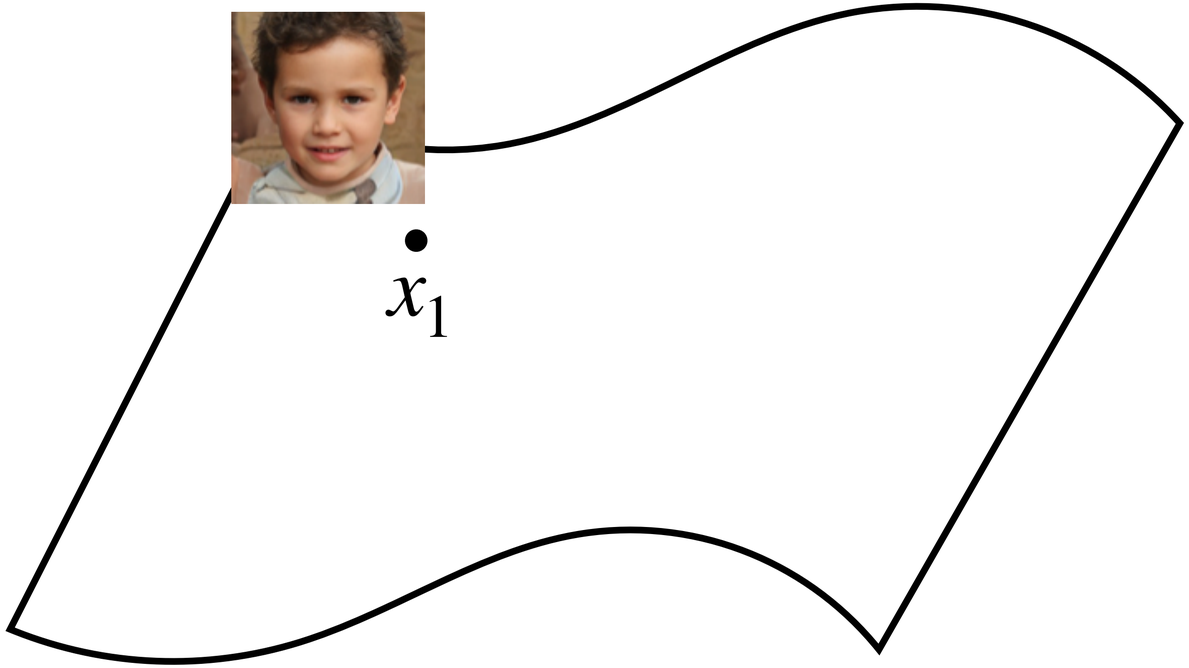
Tractable Tractable intractable

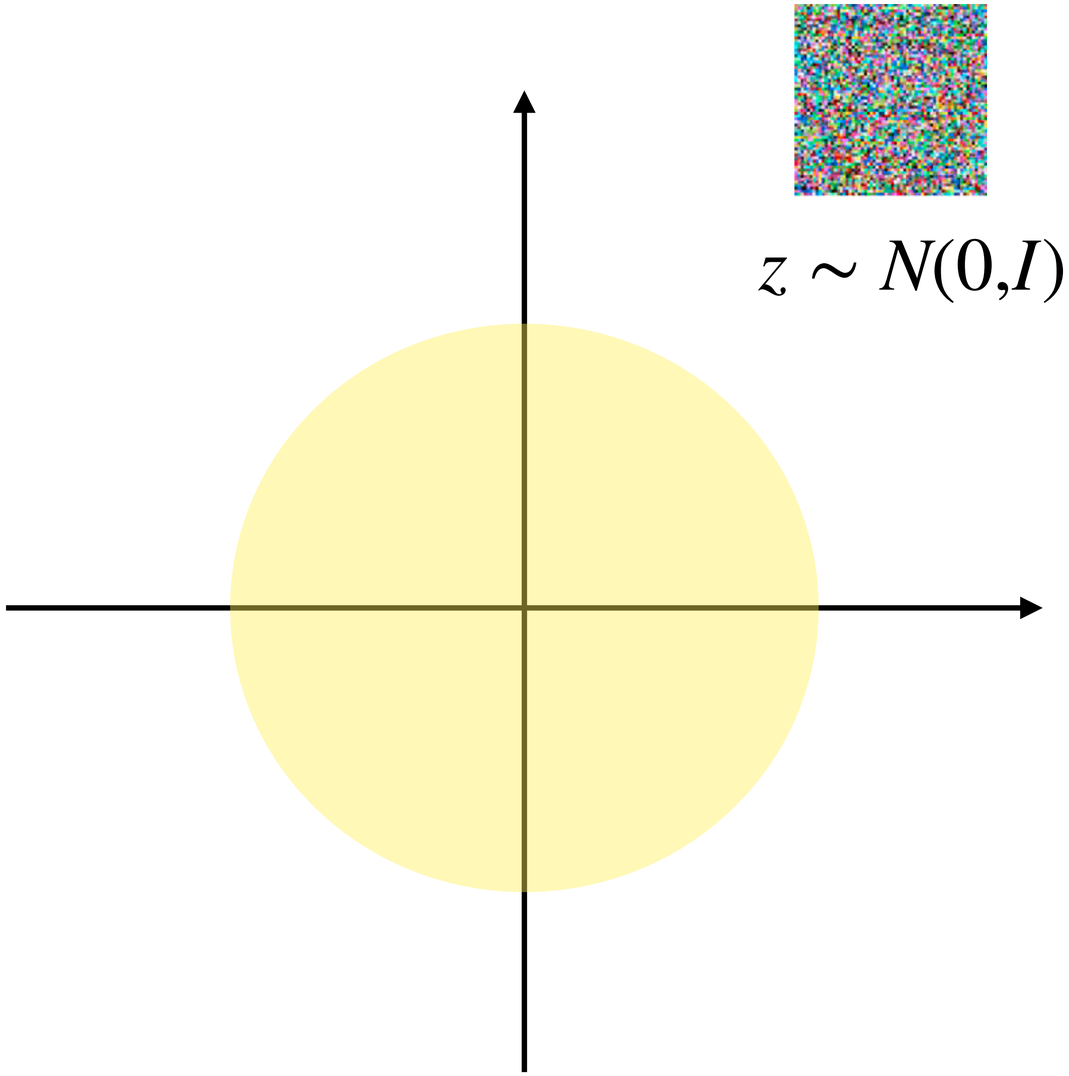
(A) : we can sample $z \sim q(\,|\,x)$ from encoder and compute $p_\theta(x\,|\,z)$ from decoder

(B) : KL divergence between Gaussian has a closed-form

(C) : we know it is just non-negative

$$\mathbb{E}_z[\log p_\theta(x|z)] - \mathrm{KL}\big(q_\phi(z|x) \| p(z)\big)$$

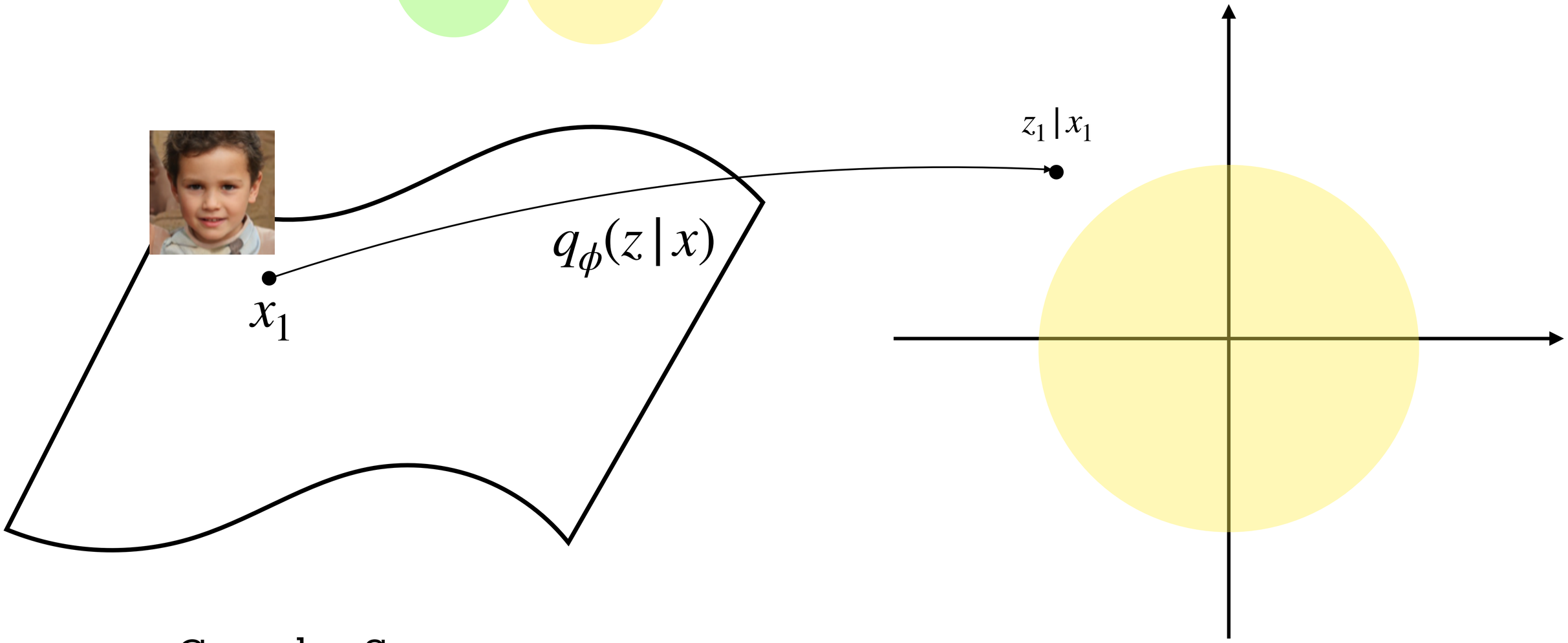




X Complex Space
= Data space

Z Simple Space
= Latent Space

$$\mathbb{E}_z[\log p_\theta(x|z)] - \mathrm{KL}\big(q_\phi(z|x) \| p(z)\big)$$

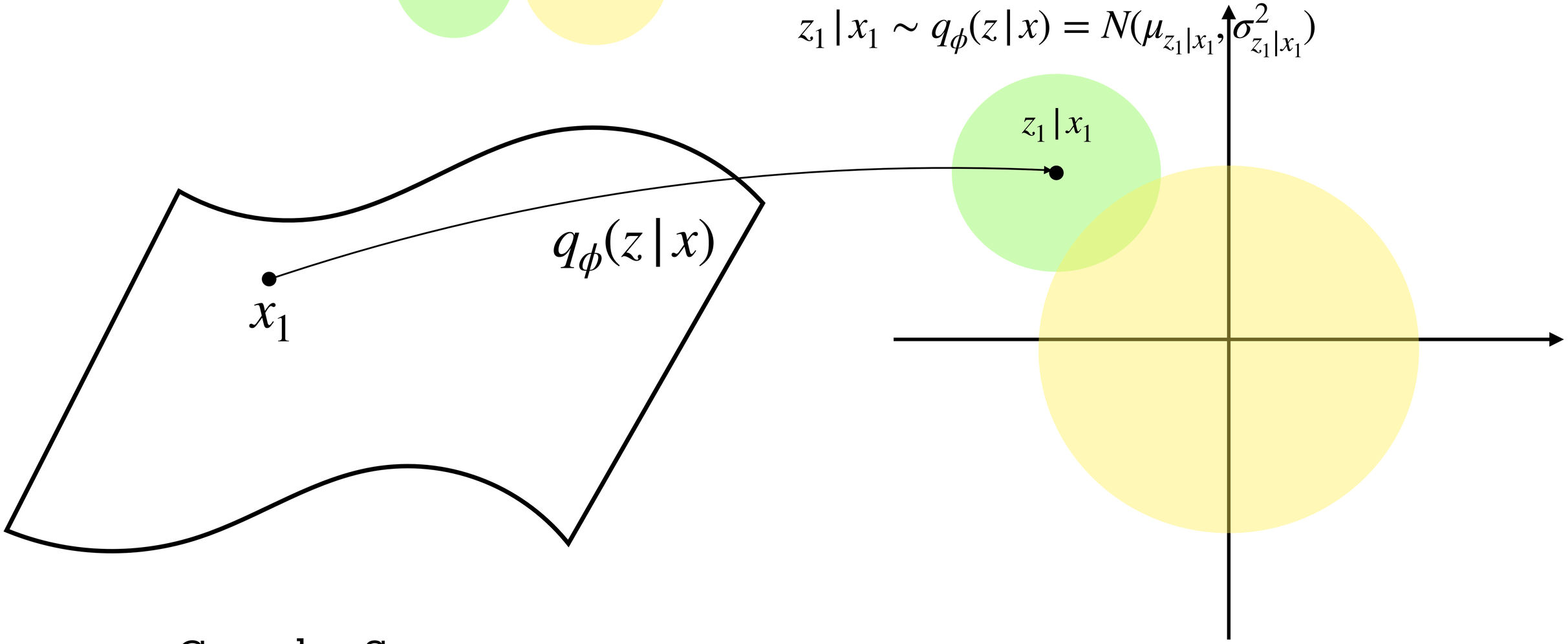


X Complex Space
= Data space

Z Simple Space
= Latent Space

$$\mathbb{E}_z[\log p_\theta(x|z)] - \mathrm{KL}(q_\phi(z|x)\|p(z))$$

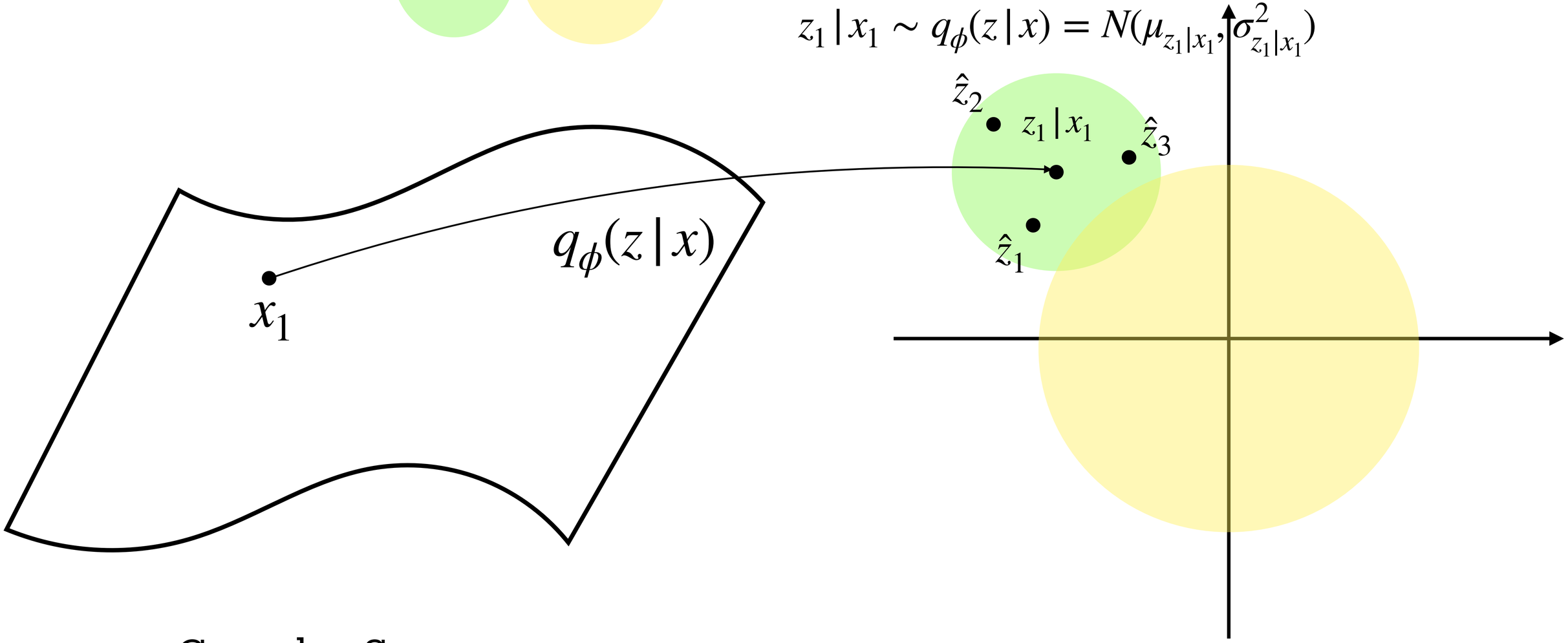
$$z_1 \,|\, x_1 \sim q_\phi(z \,|\, x) = N(\mu_{z_1|x_1}, \sigma^2_{z_1|x_1})$$



X Complex Space
= Data space

Z Simple Space
= Latent Space

$$\mathbb{E}_z[\log p_\theta(x|z)] - \mathrm{KL}(q_\phi(z|x)\|p(z))$$

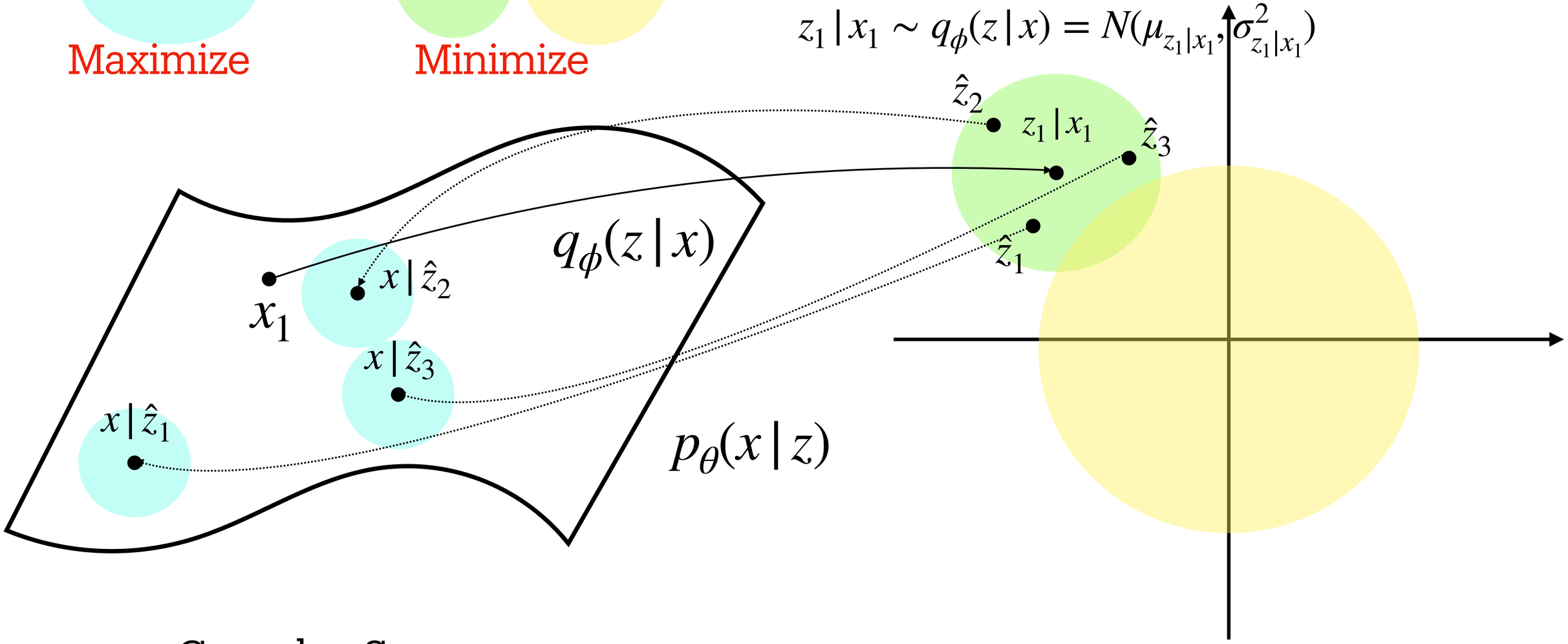


X Complex Space
= Data space

Z Simple Space
= Latent Space

$$\mathbb{E}_z[\log p_\theta(x|z)] - \mathrm{KL}(q_\phi(z|x)\|p(z))$$

Maximize Minimize

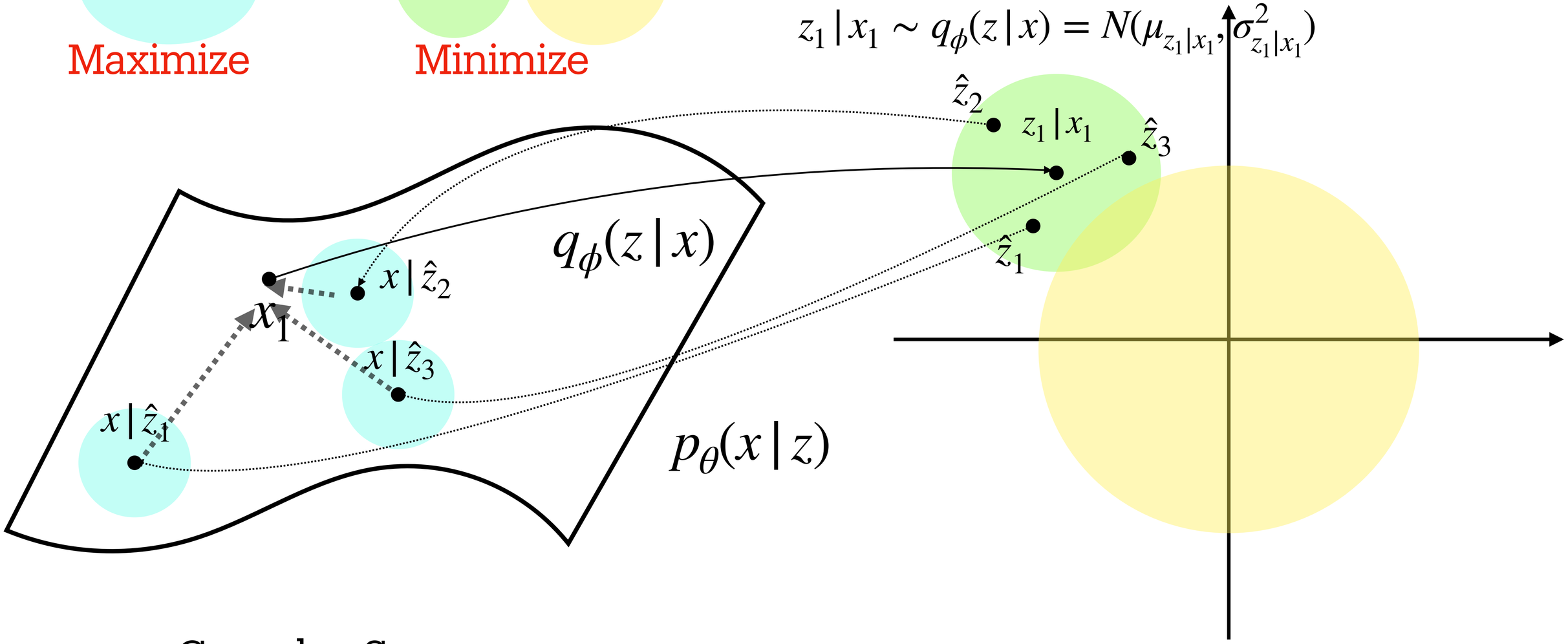


X Complex Space
= Data space

Z Simple Space
= Latent Space

$$\mathbb{E}_z[\log p_\theta(x|z)] - \mathrm{KL}(q_\phi(z|x)\|p(z))$$

Maximize Minimize

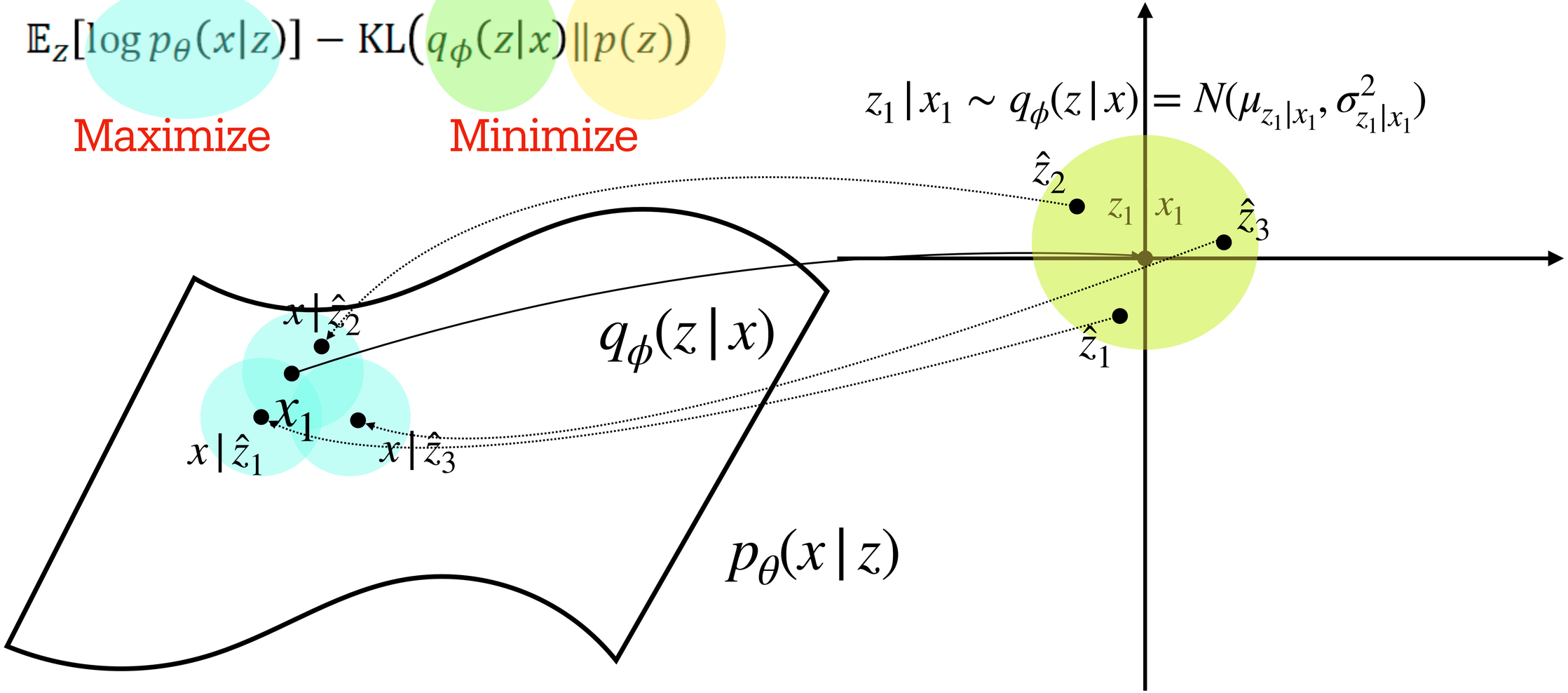


X Complex Space
= Data space

Z Simple Space
= Latent Space

$$\mathbb{E}_z[\log p_\theta(x|z)] - \mathrm{KL}(q_\phi(z|x)\|p(z))$$

Maximize Minimize



X Complex Space
= Data space

Z Simple Space
= Latent Space